

Question 1: Missing Value Imputation (Coding Case)

Case Study

You are given a student dataset containing missing values in the **Age** and **Marks** columns.

Task

1. Load the dataset into a Pandas DataFrame.
2. Identify missing values in each column.
3. Replace missing values using an appropriate **imputation strategy**.
4. Display the cleaned dataset.

Dataset

```
data = {  
    "Age": [18, 19, None, 20, 21],  
    "Marks": [85, None, 78, 90, None]  
}
```

Question 2: Categorical Data Handling (Coding Case)

Case Study

A college collects student gender data, but some values are missing.

Task

1. Handle missing values in the **Gender** column.
2. Encode the Gender column into numerical format.
3. Display the transformed dataset.

Dataset

```
data = {  
    "Gender": ["Male", "Female", None, "Male", "Female"]  
}
```

Question 3: Train-Test Sampling (Coding Case)

Case Study

You are building a model to predict student performance. The dataset must be split into training and testing data.

Task

1. Separate features and target variable.
2. Split the dataset into **70% training** and **30% testing** data.
3. Display the shapes of all four datasets.

Dataset

```
data = {  
    "Study_Hours": [2, 4, 6, 8, 10],  
    "Marks": [50, 60, 70, 85, 95]  
}
```

question- **case study-**

Student Tasks (All Coding Based)

Task 1: Load the Dataset

- Read the CSV file using `pandas.read_csv()`
- Display the first 5 rows

Task 2: Data Inspection

- Display dataset information
- Identify missing values in each column

Task 3: Handle Missing Values

- Replace missing values in:
 - Age and Final_Marks using **mean imputation**
 - Gender using **most frequent value**

- Attendance_Percentage using **mean imputation**

Task 4: Encode Categorical Data

- Convert the Gender column into numerical form

Task 5: Feature Selection

- Select:
 - Age
 - Gender
 - Attendance_Percentage
- Set Final_Marks as the target variable

Task 6: Data Discretization

- Discretize the Age column into **3 uniform bins**
- Store the result in a new column named Age_Group

Task 7: Train–Test Split

- Split the data into:
 - **70% training**
 - **30% testing**
- Use a fixed random state